

Sam Kouteili
ROSE Lab x EPI • Yale University

The Yale EPI ranks 180 countries on 58 indicators, but many suffer patchy coverage, heavy imputation, and reporting lags. We introduce a two-stage multi-agent pipeline that autonomously discovers and validates data proxies — alternative signals for hard-to-measure indicators. Across five pilot indicators it generated 48 hypotheses and validated 31 (76 %).

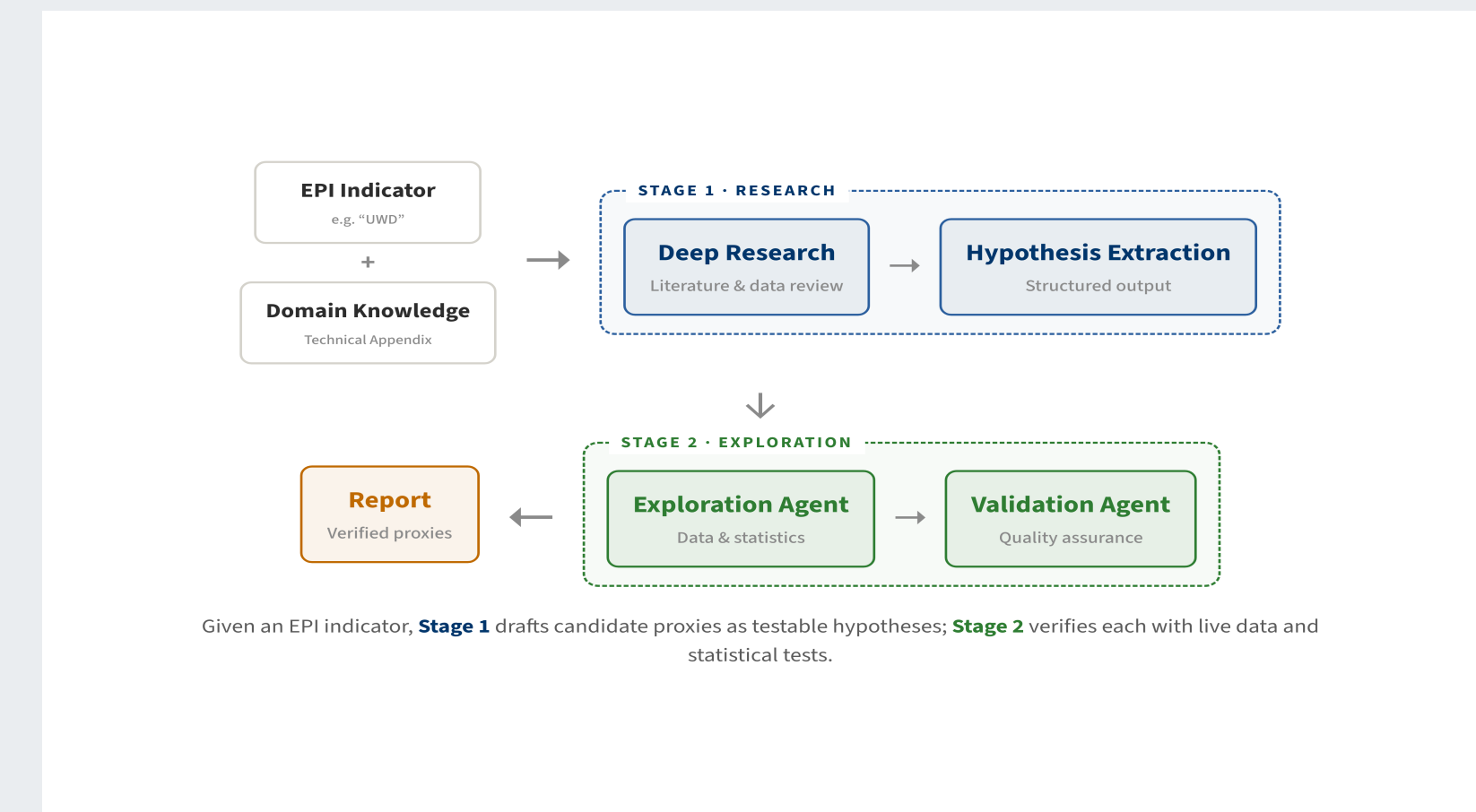
The EPI aspires to rank 180 countries on environmental performance, but cannot measure what it cannot observe:

- Waste Recovery Rate: 56 of 180 countries imputed.
- Wastewater indicators: a single year (2015) for most countries.
- Pesticide Risk: only two global time points (2015, 2018).

When official statistics are thin, the index loses its grip on the reality it aims to describe. Validated proxies offer a way in.

Can LLM agents autonomously discover, formalize, and statistically validate proxies for hard-to-measure EPI indicators — from literature to live data to verdicts — without a human in the loop?

Each candidate is formalized as $h = \psi(c, v, r)$ — context, variables, relationship — following DiscoveryBench. Stage 2 fetches data through nine public sources (World Bank, WHO, UN Comtrade, NASA POWER, OpenAQ, Earth Engine, GDELT, ...).



Five pilot indicators (UWD, WRR, SPI, OEB, PHL):

- 48 hypotheses generated; 41 verified
- 9 confirmed | 22 partially | 8 rejected | 2 inconclusive

Strongest validated proxies (Pearson r):

- Basic drinking-water access $\leftrightarrow$ UWD: $r = -0.814$
- Open-defecation rate $\leftrightarrow$ UWD: $r = +0.723$
- Terrestrial protected-area % $\leftrightarrow$ SPI: $r = +0.604$
- Fossil-fuel electricity % $\leftrightarrow$ WRR: $r = -0.409$

48 hypotheses generated

31 confirmed or partially confirmed

$r = -0.81$ strongest validated proxy (UWD)

Pilot indicators: UWD · WRR · SPI · OEB · PHL

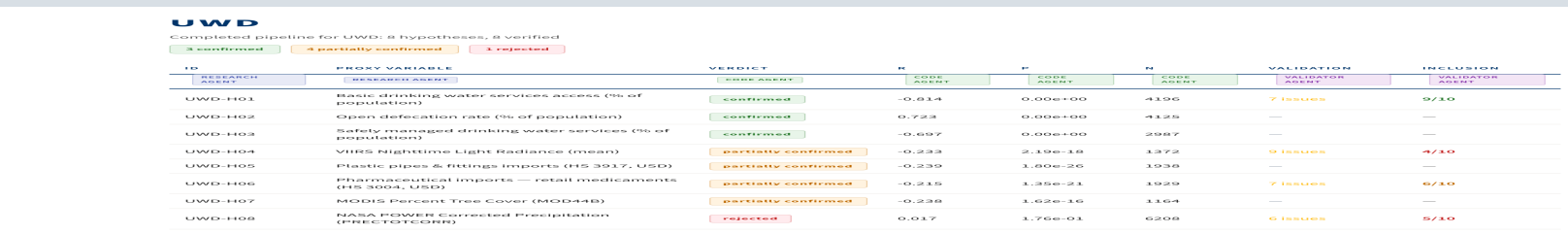


Fig 2. Per-hypothesis dashboard (UWD indicator)

A general-purpose agentic pipeline can discover and validate scientific hypotheses without a human in the loop. The recipe transfers beyond EPI to any domain asking “what else correlates with this?”

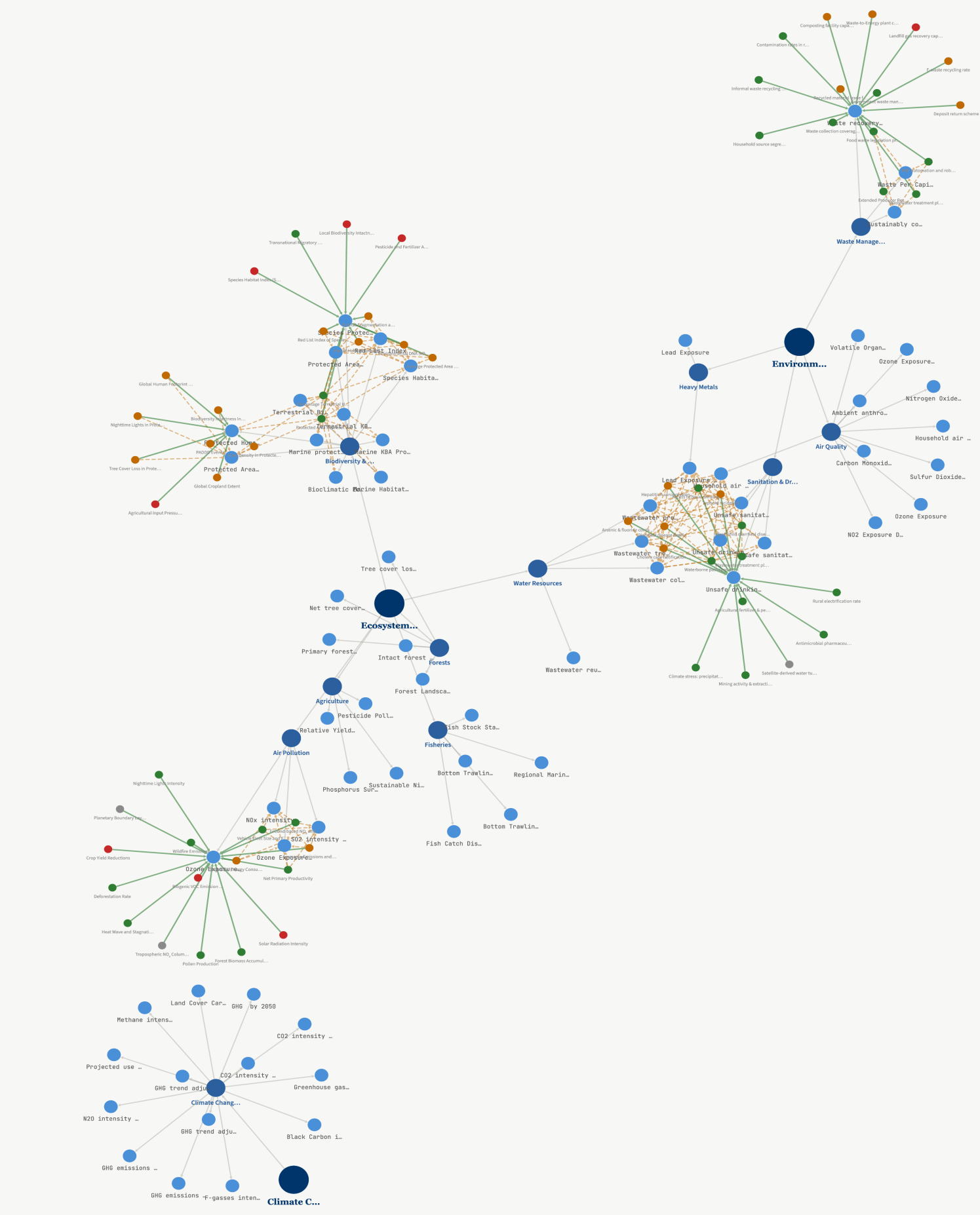


Fig 1. Knowledge graph of verified proxy relationships

Block, S. et al. (2024). 2024 EPI. Yale CELP.

Majumder, B. P. et al. (2024). DiscoveryBench. arXiv:2407.01725.

Anthropic (2025). Claude Code SDK.

Google DeepMind (2025). Gemini Deep Research API.

Sam Kouteili
samkouteili@gmail.com
github.com/SamKouteili
ROSE · Yale University